



Practical T_1 Analysis for Quadrupolar Solid-State NMR Experiments

YeonJun Choi¹, Rui Li¹, and Joseph H. Ross Jr.^{1,2}

¹ Dept. of Physics and Astronomy, Texas A&M University; ² Dept. of Materials Science and Engineering, Texas A&M University

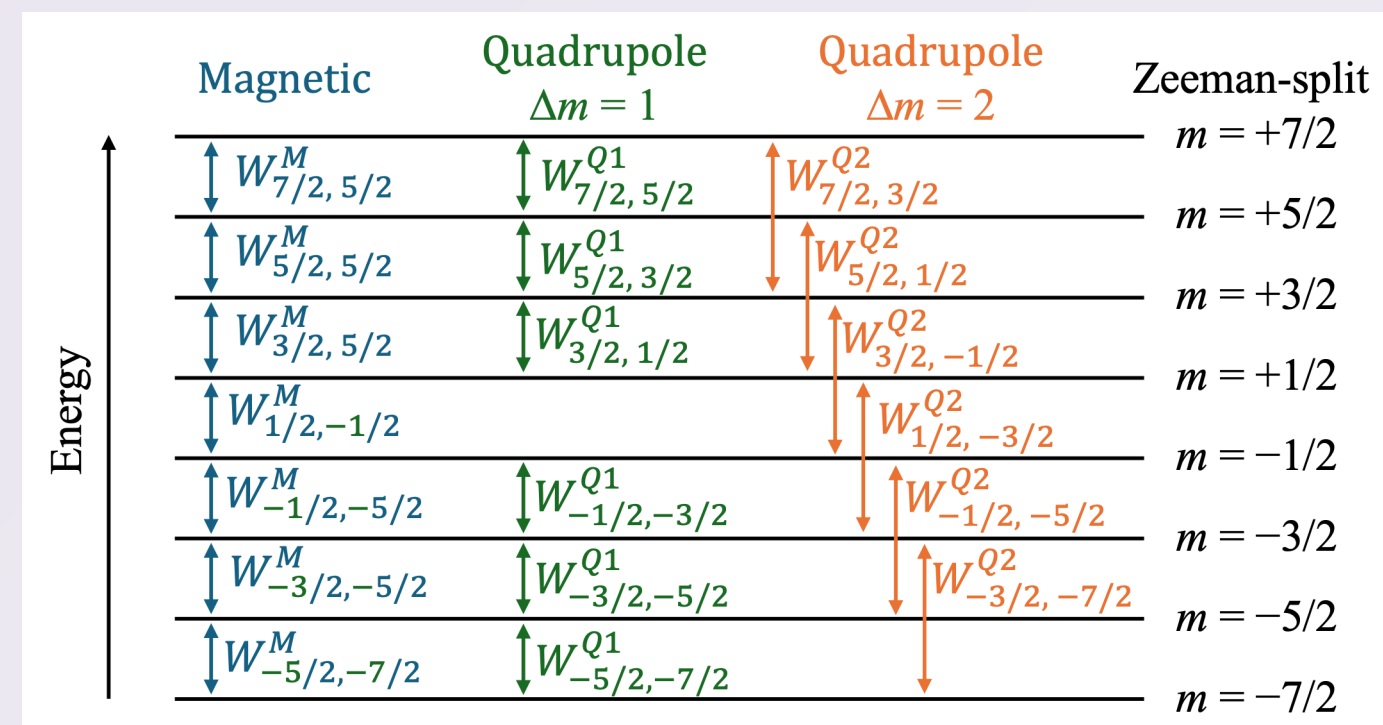
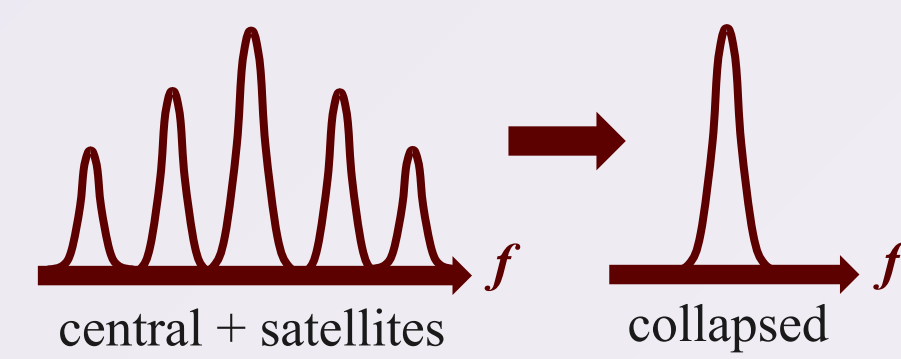


1. Motivation

- What is NMR?** Nuclear magnetic resonance probes atomic-scale structure and dynamics through nuclear spins in a magnetic field.
- Why study relaxation?** The recovery time T_1 is sensitive to atomic motion, electronic structure, and defects $\rightarrow$ a powerful, non-destructive probe of materials.
- What is T_1 ?** It measures how nuclear spins return to equilibrium after an RF pulse.
- Why is the analysis difficult?** Quadrupolar nuclei have several possible relaxation pathways.
- These pathways often produce recovery curves that are not single-exponential.
- Goal:** Find simple and reliable ways to obtain useful T_1 values from complex recovery curves.

2. Model and Method

The NMR line has a central transition plus **quadrupole satellites** (sketch at right). When quadrupole splitting vanishes, the satellites **collapse** onto the central line, reshaping the recovery.



Energy diagram for $I = 7/2$ case, showing magnetic and quadrupole rates W_{mn}^M , and W_{mn}^Q .

- Half-integer spins considered: $I = 5/2, 7/2$, and $9/2$.
- W^M : relaxation caused by fluctuating magnetic fields.
- W_1^Q : quadrupolar relaxation with $\Delta m = 1$.
- W_2^Q : quadrupolar relaxation with $\Delta m = 2$.
- Recovery curves are calculated numerically after selective saturation.¹⁻³

Relaxation Quantities

$$\Delta S(t) = S_\infty - S(t)$$

$S(t)$: observed NMR signal at recovery time t .
 S_∞ : Fully relaxed equilibrium signal.

Initial rate

$$\left. \frac{d(\Delta S)}{dt} \right|_{t=0} = -\frac{\Delta S}{T_1^{\text{init}}}$$

Exact initial recovery rate immediately after saturation⁴; linear in W^M , W_1^Q , and W_2^Q .

Single-exponential fit

$$S(t) - S(\infty) \propto e^{-t/T_1^{\text{eff}}}$$

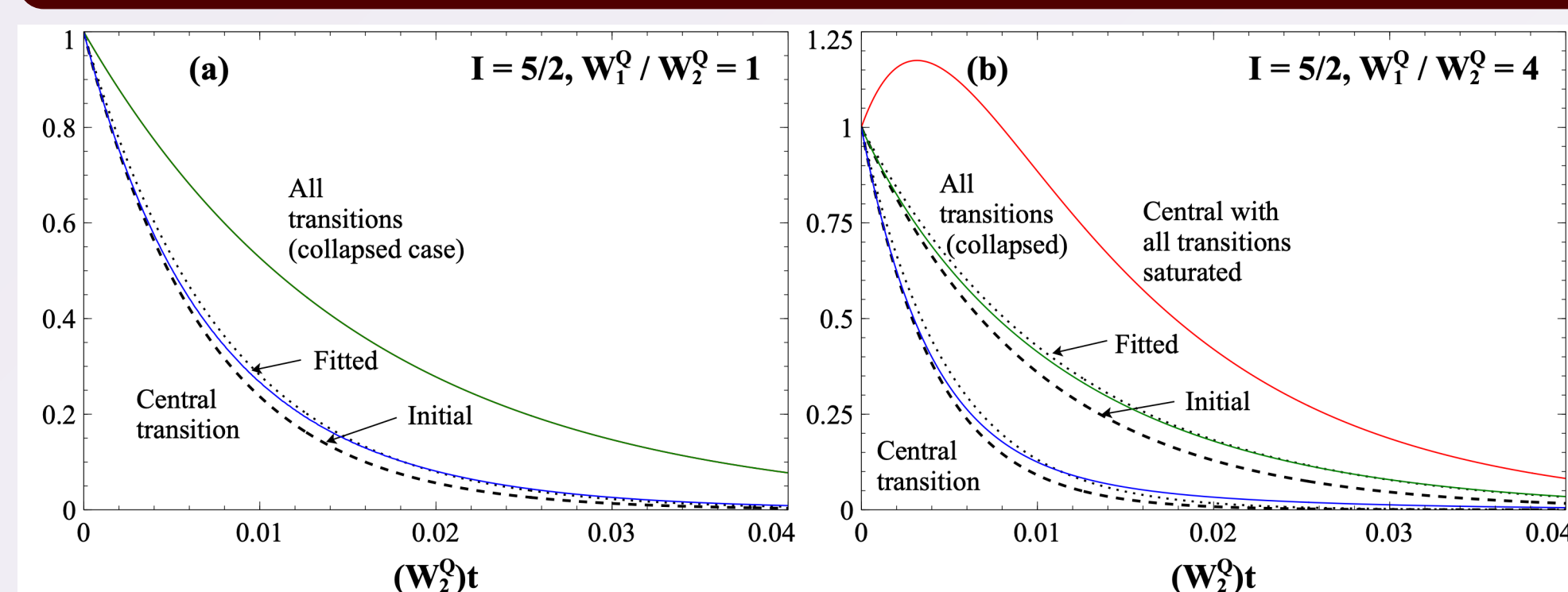
Effective rate representing the main recovery region (“knee” of the curve).

Stretched-exponential fit

$$\Delta S(t) \propto e^{-(t/T_1^{\text{str}})^\beta}$$

Fit over the full recovery curve; $\beta = 1$ is single-exponential; smaller β indicates stronger non-exponential behavior.⁵

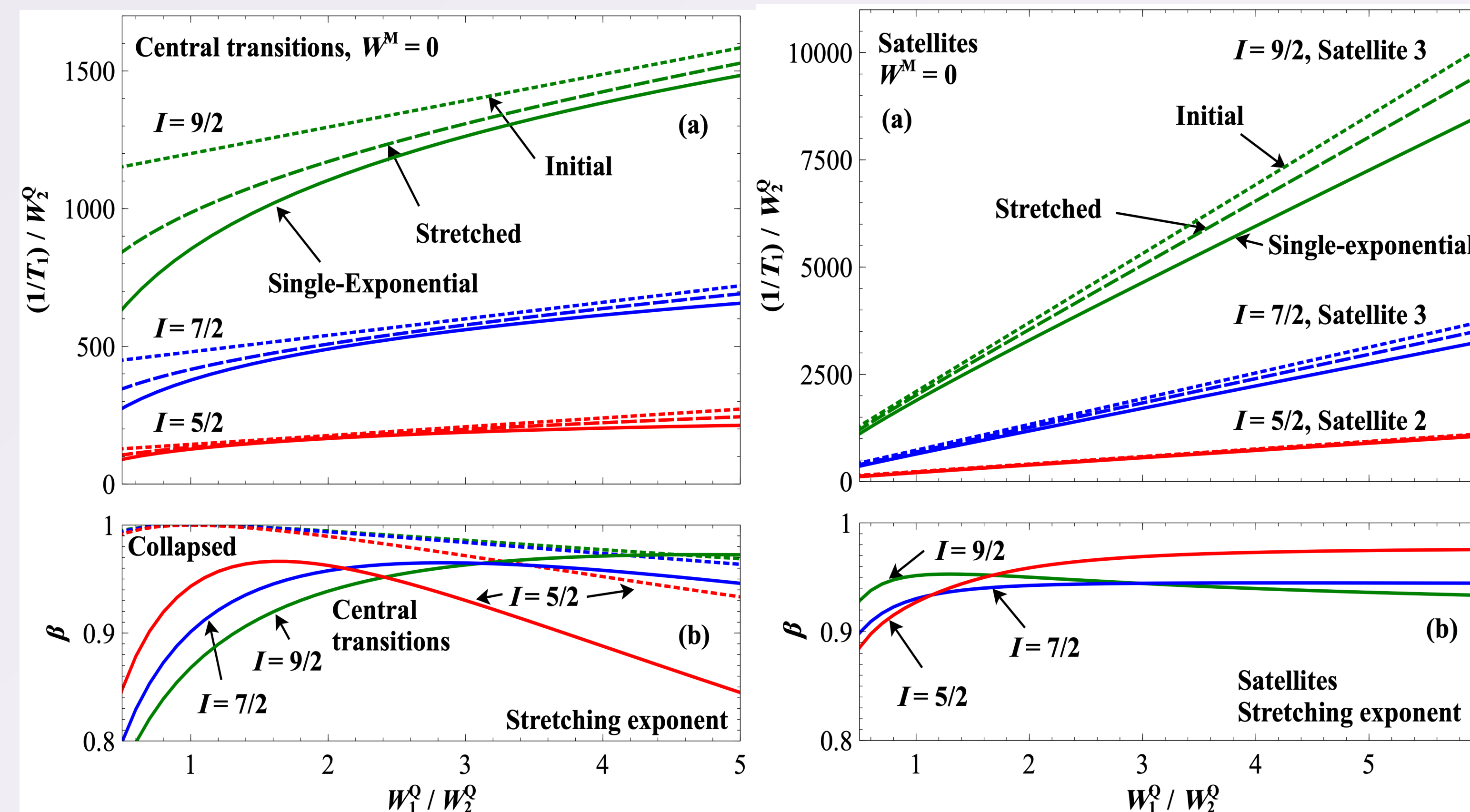
3. Recovery Curves for Different Saturation Conditions



Solid: numerical; dotted: single-exp. fit; dashed: initial-slope exponential.

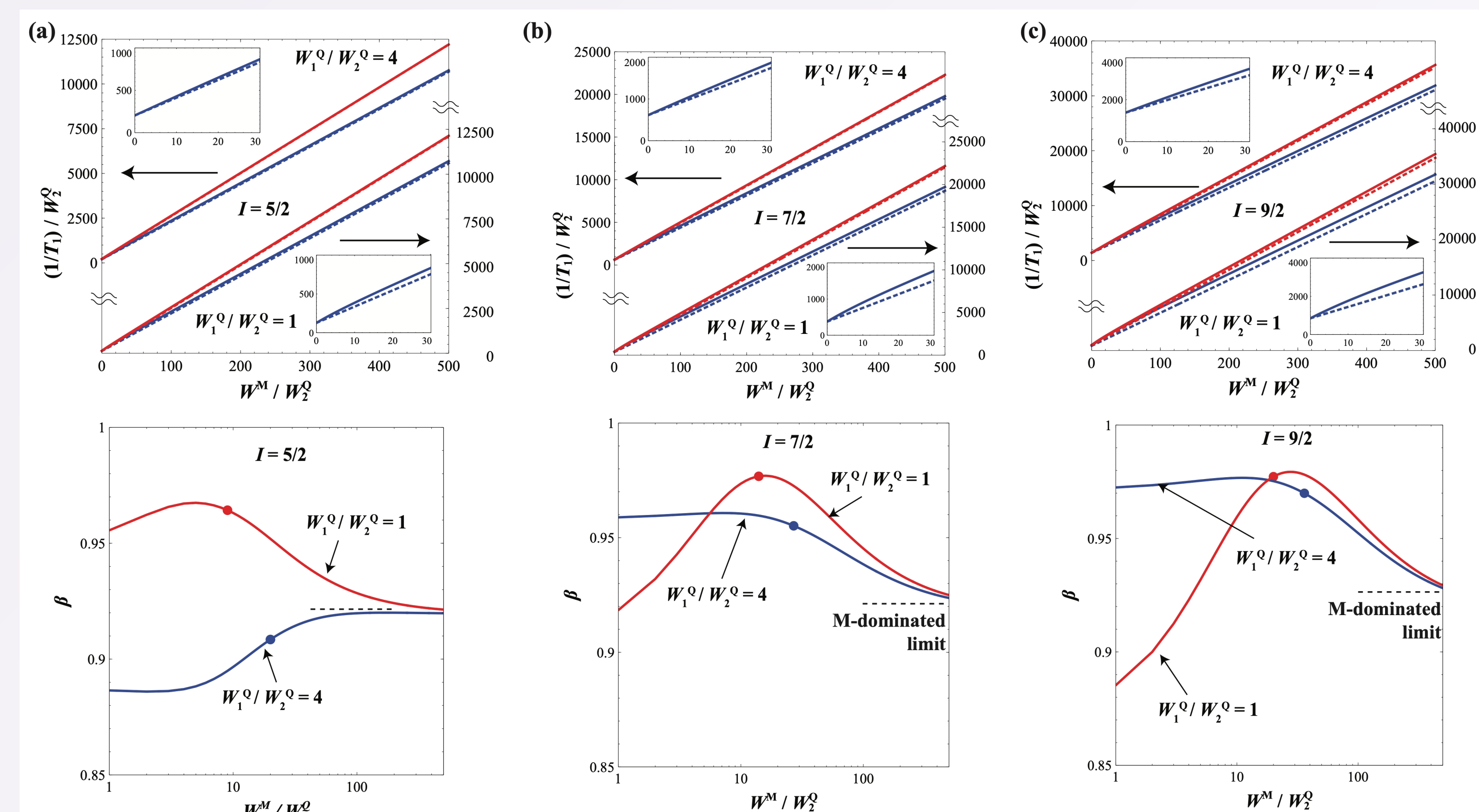
- Uniform saturation:** single-exponential when $W_1^Q = W_2^Q$, but central-only observation turns **non-monotonic** at $W_1^Q/W_2^Q = 4$.
- Collapsed satellites:** nearly single-exponential, but $T_1^{\text{eff}} \neq T_1^{\text{init}}$ in general.⁶

4. Quadrupole-only Central and Satellite Transitions



- Central transition:** stays near single-exponential over $1 \leq W_1^Q/W_2^Q \leq 4$ ($\beta \gtrsim 0.85$ for all I).
- Deviations grow as $W_1^Q/W_2^Q \rightarrow 0$ ($\Delta m = 2$ does not connect all levels).
- Satellites:** closer to single-exponential and strongly W_1^Q/W_2^Q -dependent; best for separating the $\Delta m = 1$ and $\Delta m = 2$ channels.

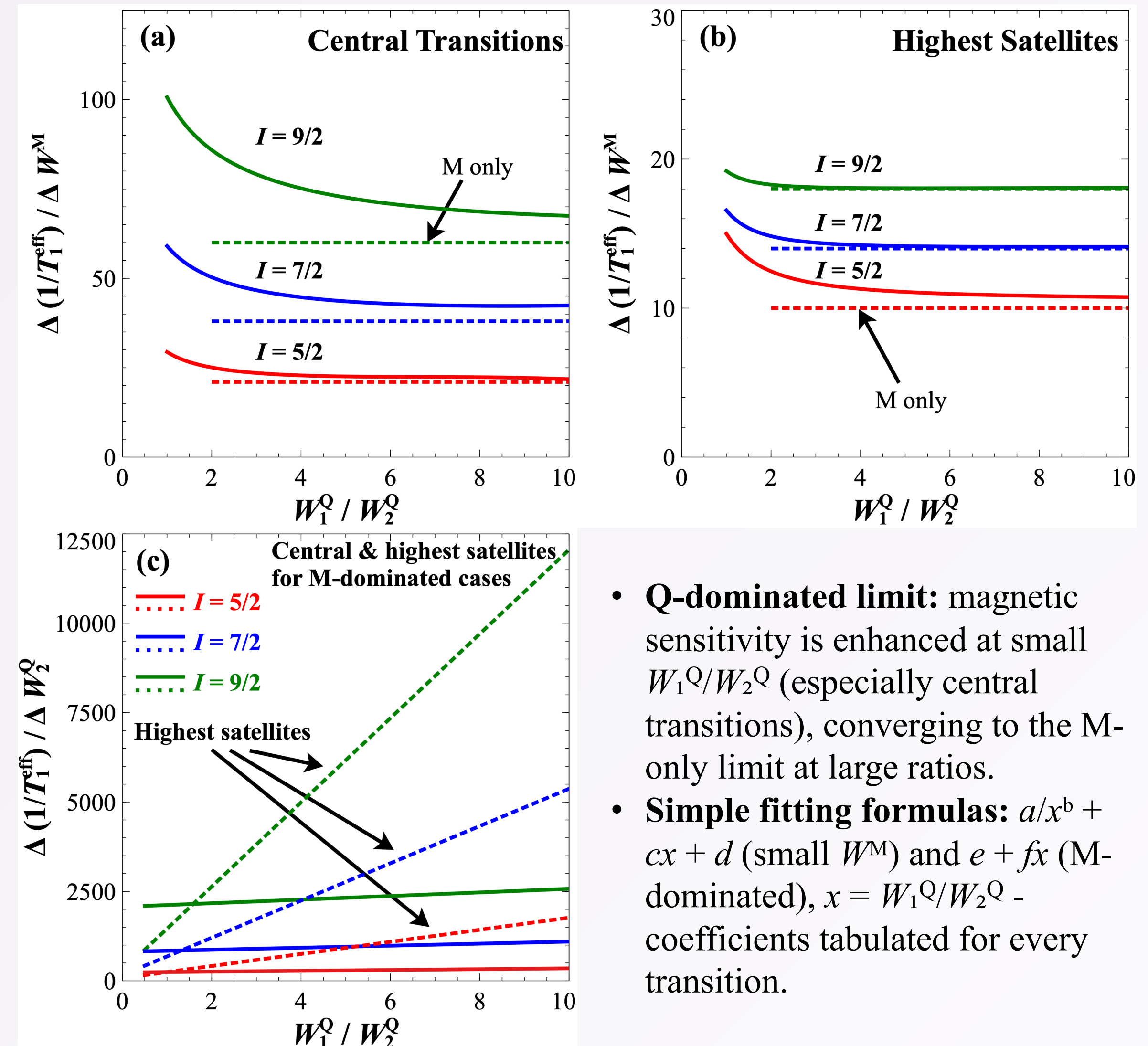
5. Combined Magnetic and Quadrupole Relaxation



Blue/red: single-/stretched-exponential fits; solid/dashed: full M+Q result / simple-sum estimate. In β panels, red/blue: $W_1^Q/W_2^Q = 1, 4$; dots: maximum deviations.

- Simple sum $1/T_{1,M} + 1/T_{1,Q}$ underestimates** the true effective rate; worst near $1/T_{1,M} \approx 1/T_{1,Q}$.
- By up to 15-25%** ($I = 5/2 \rightarrow 9/2$) at $W_1^Q/W_2^Q = 1$; only 4-10% at $W_1^Q/W_2^Q = 4$.
- Stretched fitting halves the discrepancy**, tracking the additive limit closely.

6. Practical Parameterization in the Q- / M- Dominated Limits



- Q-dominated limit:** magnetic sensitivity is enhanced at small W_1^Q/W_2^Q (especially central transitions), converging to the M-only limit at large ratios.
- Simple fitting formulas:** $a/x^b + cx + d$ (small W^M) and $e + fx$ (M-dominated), $x = W_1^Q/W_2^Q$ - coefficients tabulated for every transition.

7. Conclusions

- Central-transition Q-only recovery stays close to single-exponential for $1 \leq W_1^Q/W_2^Q \leq 4$; simple fits give reliable T_1 values.
- Satellite transitions are closer to single-exponential and depend strongly on W_1^Q/W_2^Q ; best suited for separating W_1^Q , W_2^Q , and W^M .
- The additive estimate $1/T_{1,M} + 1/T_{1,Q}$ underestimates the true rate; stretched-exponential fits track the additive limit more closely.
- Fitted parameters are tabulated for direct experimental use; the approach has been applied to NiTiHf/NiTiZr shape-memory and strain-glass dynamics.⁷

References

- A. Abragam, *The Principles of Nuclear Magnetism* (Oxford Univ. Press, 1961).
- C. P. Slichter, *Principles of Magnetic Resonance*, 3rd ed. (Springer, 1990).
- D. E. MacLaughlin, J. D. Williamson, and J. Butterworth, Phys. Rev. B **4**, 60 (1971).
- T. Rega, J. Phys.: Condens. Matter **3**, 1871 (1991).
- D. C. Johnston, Phys. Rev. B **74**, 184430 (2006).
- B. Halle and H. Wennerström, J. Magn. Reson. **44**, 89 (1981).
- R. Li *et al.*, Acta Mater. **294**, 121099 (2025).